

Data Structures Lab 3

Task 1

Source Code:

```
#include <iostream>
using namespace std;

void recordAltitude(int*& altitudes, int& capacity, int& currentCount, int newAltitude){
    if(currentCount >= capacity)
    {
        capacity = (capacity == 0) ? 1 : capacity * 2;
        int* newPtr = new int[capacity];
        for(int i = 0; i < currentCount; i++) newPtr[i] = altitudes[i];
        delete[] altitudes;
        altitudes = newPtr;
    }
    altitudes[currentCount++] = newAltitude;
}

int main(){
    int size = 0, count = 0, newRead = 0, choice = 0;
    cout << "Enter the number of readings you want to store:";
    cin >> size;
    int* ptr = new int[size];
    cout << "Enter number of readings you want to store right now:";
    cin >> count;
    if(count > size) count = size;
    for (int i = 0; i < count; i++) cin >> ptr[i];
    cout << "Do you want to enter another reading?\n1. Yes\n2. No\n";
    cin >> choice;
    switch(choice)
    {
        case 1:
            cout << "Enter the new Reading:";
            cin >> newRead;
            recordAltitude(ptr, size, count, newRead);
            break;
        case 2:
```

```

                break;
            default:
                break;
        }
        cout << "\n===All Readings of the Drone===\n";
        for (int i = 0; i < count; i++) cout << ptr[i] << " " << endl;
        delete[] ptr;
        return 0;
    }
}

```

Screenshot:

```

muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$ g++ Task1.cpp
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$ ./a.out
Enter the number of readings you want to store:4
Enter number of readings you want to store right now:4
10 20 30 40
Do you want to enter another reading?
1. Yes
2. No
1
Enter the new Reading:50

===All Readings of the Drone===
10
20
30
40
50
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$

```

Task 2

Source Code:

```
#include <iostream>
using namespace std;

void purgeErrors(int*& readings, int& count){
    int trace = 0;
    for (int i = 0; i < count; i++) if(readings[i] == -999) trace++;

    int newCount = count - trace;
    int* newPtr = new int[newCount];

    int k = 0;
    for (int i = 0; i < count; i++)
    {
        if(readings[i] == -999) continue;
        newPtr[k++] = readings[i];
    }

    delete[] readings;
    readings = newPtr;
    count = newCount;
}

int main(){
    int size = 0;
    cout << "Enter the number of readings you want to store:";
    cin >> size;
    cout << "Enter the readings: ";
    int* ptr = new int [size];
    for (int i = 0; i < size; i++) cin >> ptr[i];
    purgeErrors(ptr, size);
}
```

```

        cout << "\n===All Readings of the Drone===\n";
        for (int i = 0; i < size; i++) cout << ptr[i] << " ";
        delete[] ptr;
        return 0;
}

```

Screenshot:

```

muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$ g++ Task2.cpp
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$ ./a.out
Enter the number of readings you want to store:5
Enter the readings: 1 -999 2 -999 3

===All Readings of the Drone===
1 2 3 muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$

```

Task 3

Source Code:

```

#include <iostream>
using namespace std;
int* mergeDatabases(int* dbA, int sizeA, int* dbB, int sizeB){
    int SIZE = sizeA + sizeB;
    int* merged = new int[SIZE];
    int i = 0, j = 0, k = 0;
    while(k < SIZE)
    {
        if(i < sizeA && j < sizeB)
        {
            if(dbA[i] < dbB[j]) merged[k++] = dbA[i++];
            else merged[k++] = dbB[j++];
        }
    }
}

```

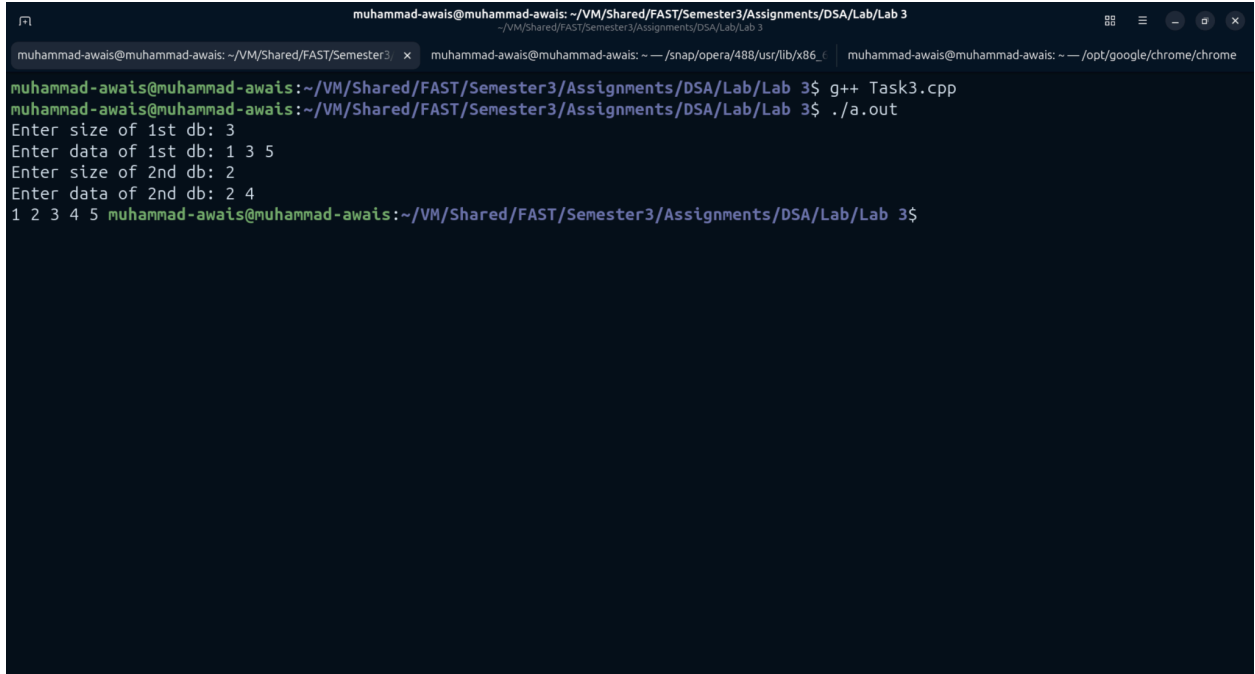
```

        }
        else if(i < sizeA)
        {
            merged[k++] = dbA[i++];
        }
        else
        {
            merged[k++] = dbB[j++];
        }
    }
    return merged;
}

int main(){
    int *ptrA, *ptrB, *mergedDB;
    int size1, size2;
    cout << "Enter size of 1st db: "; cin >> size1;
    ptrA = new int[size1];
    cout << "Enter data of 1st db: ";
    for (int i = 0; i < size1; i++) cin >> ptrA[i];
    cout << "Enter size of 2nd db: "; cin >> size2;
    ptrB = new int[size2];
    cout << "Enter data of 2nd db: ";
    for (int i = 0; i < size2; i++) cin >> ptrB[i];
    mergedDB = mergeDatabases(ptrA, size1, ptrB, size2);
    delete[] ptrA;
    delete[] ptrB;
    for(int i = 0; i < size1+size2; i++) cout << mergedDB[i] << " ";
    delete[] mergedDB;
    return 0;
}

```

Screenshot:



```
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$ g++ Task3.cpp
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$ ./a.out
Enter size of 1st db: 3
Enter data of 1st db: 1 3 5
Enter size of 2nd db: 2
Enter data of 2nd db: 2 4
1 2 3 4 5 muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab 3$
```